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SCHOOL FUTURE TECH PROGRAMM

Faculty Name :

Course Duration	: 5 Months (150 Days)
Classes	: Offline & Online
Daily Teaching	: 3 Hours
Total Hours	: 300 Hours

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LEVEL 1 – JUNIOR INNOVATORS (5th–6th Standard)

MODULE 1 – COMPUTER BASICS (Classes 1–20)

Class 5 — Input and Output Devices (Complete Notes)

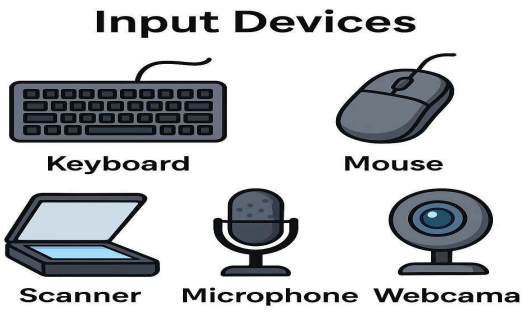
Complete Theory & Practical Notes

Learning Objectives:-

By the end of this class, students will be able to:-

- Understand the meaning of Input and Output devices.
- Identify common input and output devices.
- Explain the function of each device.
- Differentiate between input and output devices.
- Complete a practical identification activity.
- Create an interactive Scratch project about input and output devices.

What are Input and Output Devices?





A computer works by receiving information, processing it, and giving results.

The devices that **send information to the computer** are called **Input Devices.**

The devices that **show or produce results from the computer** are called **Output Devices.**

Computer Working Process

Input ➜ Processing (CPU) ➜ Output

Example:-

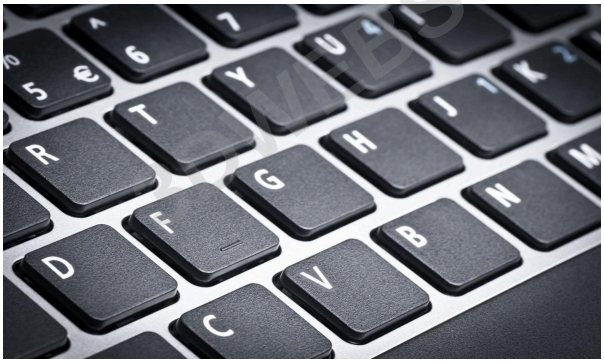
You type your name using a keyboard ***(Input)*** → The CPU processes it → Your name appears on the monitor ***(Output).***

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What is an Input Device?

An ***Input Device*** is a device used to enter data and instructions into a computer.

Common Input Devices







Keyboard:-

- Used for typing letters and numbers.
- Most common input device.

Mouse:-

- Used to click, drag, and select items.

Microphone:-

- Used to record voices.
- Used in online classes and voice assistants.

Scanner:-

- Converts paper documents into digital files.

Webcam:-

- Captures photos and videos.
- Used in video calls.

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What is an Output Device?

An ***Output Device*** displays or produces the results processed by the computer.

Common Output Devices





Monitor:-

Displays:-

- Pictures
- Videos
- Text
- Games

Speakers:-

Produce sound such as:-

- Music
- Videos
- Online classes

Printer:-

Prints:-

- Photos
- Documents
- Assignments

Projector:-

Displays the computer screen on a large wall or screen.

Headphones:-

Allow one person to hear computer audio privately.

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Easy Memory Trick

Device Type	*What it Does*
Input Device	Sends data **to** the computer
Output Device	Receives results **from** the computer

Examples

Input Devices	Output Devices
Keyboard	Monitor
Mouse	Speakers
Scanner	Printer
Microphone	Projector
Webcam	Headphones

Device Classification:-

Device	Input	Output
Keyboard	✅	❌
Mouse	✅	❌
Scanner	✅	❌
Monitor	❌	✅
Printer	❌	✅
Speaker	❌	✅
Microphone	✅	❌
Projector	❌	✅

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Number of Devices by Category

Input vs Output Devices

Examples of devices covered in today's lesson.

0246Input DevicesOutput Devices

Classroom Practical Activity:-

Activity 1 – Identify the Device

The teacher places different devices on a table or shows pictures.

Students identify:

- Device Name
- Input or Output?
- What is it used for?

Example Table:-

Device	*Type*	*Use*
Keyboard	Input	Typing
Mouse	Input	Clicking
Monitor	Output	Display
Speaker	Output	Sound
Printer	Output	Printing

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Activity 2 – Input or Output Game

The teacher shows one device at a time.

Students raise:-

🟢 Green Card = Input Device

🔵 Blue Card = Output Device

Examples:-

- Keyboard → 🟢
- Monitor → 🔵
- Mouse → 🟢
- Printer → 🔵
- Scanner → 🟢
- Speaker → 🔵

Activity 3 – Draw and Label

Students draw:-

- One Input Device
- One Output Device

Write:-

- Device Name
- Device Type
- One Use

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Practical Coding Activity (Scratch)

Project Title:-

"Input or Output?" Interactive Learning Game

Objective

Students create a Scratch project where clicking a device tells whether it is an Input or Output device.

Step 1

Open Scratch.

Create a New Project.

Delete the default Cat sprite.

Step 2

Add Sprites

- Keyboard
- Mouse
- Monitor
- Speaker
- Printer
- Scanner

Add a classroom or computer lab background.

Step 3

Programming

Green Flag

when green flag clicked
say "Click a device to learn whether it is Input or Output!" for 3 seconds

Keyboard Sprite

when this sprite clicked
say "Keyboard is an INPUT device." for 3 seconds

Mouse Sprite

when this sprite clicked
say "Mouse is an INPUT device." for 3 seconds

Monitor Sprite

when this sprite clicked
say "Monitor is an OUTPUT device." for 3 seconds

Speaker Sprite

when this sprite clicked
say "Speaker is an OUTPUT device." for 3 seconds

Printer Sprite

when this sprite clicked
say "Printer is an OUTPUT device." for 3 seconds

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Scanner Sprite

when this sprite clicked
say "Scanner is an INPUT device." for 3 seconds

Advanced Scratch Challenge

Create an interactive quiz.

Example:-

ask "Which device is used for typing?" and wait

If answer = Keyboard

Then

change Score by 1
say "Correct!" for 2 seconds

Else

say "Try Again!" for 2 seconds

Add More Questions

- Which device prints documents?
- Which device records your voice?
- Which device displays videos?
- Which device scans photos?
- Which device produces sound?

Display the total score at the end.

AI Connection

Discuss with students:-

- Which input devices help AI understand us? (Microphone, Webcam)
- Which output devices help AI communicate with us? (Monitor, Speakers)
- Can a voice assistant work without a microphone?
- Why are input and output devices important in robotics?

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Fun Quiz:-

1. Which is an Input Device?

- A) Monitor
- ***B) Keyboard***
- C) Speaker

2. Which device prints documents?

- A) Mouse
- B) Scanner
- ***C) Printer***

3. Which device displays pictures?

- A) Keyboard
- ***B) Monitor***
- C) Microphone

4. Which device records your voice?

- A) Printer
- ***B) Microphone***
- C) Speaker

5. Which device produces sound?

- A) Mouse
- B) Scanner
- ***C) Speakers***

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Homework:-

1. Draw ***five input devices*** and ***five output devices.***
2. Write one use of each device.
3. Create a comparison table of Input and Output devices.
4. Improve your Scratch project by adding animations, sounds, and a score variable.
5. Observe the computer at home or school and identify all the input and output devices connected to it.

Learning Outcomes

After completing this lesson, students will be able to:-

- Explain the difference between ***Input*** and ***Output*** devices.
- Identify common input and output devices correctly.
- Describe the purpose of each device.
- Participate confidently in practical identification activities.
- Create an interactive Scratch game to classify computer devices.
- Understand how input and output devices work together with the CPU to complete computer tasks.

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